NB-280: Gas phase calibration of O2 robust oxygen sensor (Firesting) in a 4 neck H2/O2 photoreactor

Date: 2025-09-10
Tags: O2 Calibration NB Firesting Degassing O2 sensor Gas phase Trace range robust oxygen sensor
Category: Prep work
Status: Done
Created by: Nadzeya Brezhneva

Reaction scheme/sample structure

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ChemDraw File (linked): /

Literature/reference experiments

Literature	/
Reproduction	/
Similar experiments	Prep work - NB-232: Gas phase calibration of O2 robust oxygen sensor (Firesting) MLB-094: Gas-phase calibration of optical oxygen sensor JSC-616: Gas-phase calibration of optical oxygen sensor

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	$Mass_{theo}$ [mg]	$Mass_{exp}$ [mg]	Molar mass [g/mol]	Density (g/ml)	Volume [ml]
milli-Q H2O	/	/	/	/	/	/	/	/	25

Irradiation Parameters

	Name
Used Set-up (select one)	Equipment - Advanced irradiation setup V1.0 I

Procedure/observations

Date	Time	Step	Observations	Pictures
10.09.2025	13:05	An Equipment - H2/O2 reactor equipped with an oval PTFE stirring bar was set in the irradiation setup.	/	/
	13:10	25 mL H_2O was transferred inside the flask using funnel, PT 100 sensor was placed near the photoreactor, running LAUDA at 25 °C.	/	/

	13:20-30	Preparation of BOLA fitting - 3 mm BOLA fitting was used for the trace robust oxygen sensor (placing the sensor above the liquid level), the other necks were closed with stoppers.	PT100 approximately the same level as the tip of O_2 sensor.	20250910_133411-assembled photoreactor with oxygen sensor.jpg
	13:40-45	Starting Pyroscience Workbench software, checking calibration parameters.	/	/
		Important note: For upper calibration point --> Factory calibrated (still ok, in green), for 0 % calibration degassing procedure was required.	/	/
	14:07	PTFE cannula was immersed inside the GL14 flask via the valve (start of purging with Ar) - start of bubbling.	/	20250910_141024-during degassing.jpg
	14:09	Start logging.	/	2025-09-10_140913_NB-280-Ch2-1-1.txt 2025-09-10_140913_NB-280-Ch2-1-1.png
	14:29	Stop logging.	/	/
	14:30	Lower calibration point was taken during calibration procedure.	/	/
	14:32	Stop degassing by removing the cannula from the flask, closing the valve.	/	/
	14:32	Start new logging.	Sealing check after degassing.	2025-09-10_143242_NB-280-Ch2-1-2.txt 2025-09-10_143242_NB-280-Ch2-1-2.png
	14:47	Stop logging.	/	/
	14:48	Save instrument configuration.	/	20250910-BOLA fitting-gas phase-4-neck photoreactor-trace oxygen robust probe-Ch2.ini
	14:50	Deassembling the setup.	/	/

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
10.09.2025	14:09	NB-280-Ch2-1-1	O_2 optical detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 2)	H_2O	2025-09-10_140913_NB-280-Ch2-1-1.txt	2025-09-10_140913_NB-280-Ch2-1-1.png	/	Degassing (purging with Ar).
	14:32	NB-280-Ch2-1-2	O_2 optical detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 2)	H_2O	2025-09-10_143242_NB-280-Ch2-1-2.txt	2025-09-10_143242_NB-280-Ch2-1-2.png	/	Sealing test after removal of cannula --> Ok. New calibration file: 20250910-BOLA fitting-gas phase-4-neck photoreactor-trace oxygen robust probe-Ch2.ini

Results

Calibration of trace robust O2 sensor in gas phase using 4-neck photoreactor was performed. New calibration file was created.

Linked experiments

- JSC-616: Gas-phase calibration of optical oxygen sensor

- MLB-094: Gas-phase calibration of optical oxygen sensor

Prep work - NB-232: Gas phase calibration of O2 robust oxygen sensor (Firesting)

Linked resources

Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 2) - CEEC II E004 C9

Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 3) - CEEC I 208 Cupboard B.1

Equipment - Advanced irradiation chamber V1.0 I

Equipment - H2/O2 reactor V1.0 - CEEC II E002

Attached files

20250910_133411-assembled-photoreactor-with-oxygen-sensor.jpg
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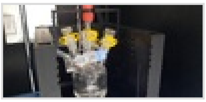

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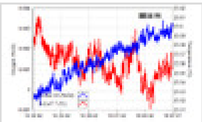

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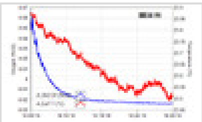

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20250910-BOLA fitting-gas phase-4-neck photoreactor-trace oxygen robust probe-Ch2.ini
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Unique eLabID: 20250910-57cd0ef891ddbe261784bceec430775a49414015
Link: https://elab.water-splitting.org/experiments.php?mode=view&id=2911